

$\epsilon_{abs} = 1.0E - 15$
 $\epsilon_{rel} = 1.0E - 13$
 $\epsilon_{func} = 1.0E - 15$

No.	$f(x)$	n	$[x_a, x_b]$	x_{root}	bisection	regula_falsi	abkk	barycentric	zhang	ridders	illinois	blendtf	itp	bdgrf	rhp	pegasus	tona748	brent	brentq	anderson	bjorck	brenth	muller	modab	chandrupatia
1	$\sin x - x/2$				1.89549426703981E+00	45	34	20	27	16	14	10	26	10	8	9	9	9	9	9	9	9	8	9	9
2					3.022915347273057E+00	46	1045*	84	60	19	18	101	32	17	22	14	102	16	15	15	52	15	52	12	12
3			$[1 + 1 \times 10^{-9}, 4 - 1 \times 10^{-9}]$		6.68873364080878E+00	45	1045*	22	26	14	16	99	26	10	12	100	9	8	8	52	11	11	11	13	
4			$[9 + 1 \times 10^{-9}, 16 - 1 \times 10^{-9}]$		1.23870165500221E+01	45	1045*	20	25	14	16	102	29	12	16	102	105	15	15	15	56	15	12	11	10
5			$[16 + 1 \times 10^{-9}, 25 - 1 \times 10^{-9}]$		1.9676000008062341E+01	45	1045*	22	40	14	16	104	26	12	16	14	104	14	13	13	56	13	11	11	10
6	$-2 \sum_{i=1}^{30} ((2i-5)^2)/(x-i^2)^3$		$[25 + 1 \times 10^{-9}, 36 - 1 \times 10^{-9}]$		2.982822732650476E+01	44	1044*	22	28	17	16	105	23	11	20	14	104	12	12	12	57	13	11	10	11
7			$[36 + 1 \times 10^{-9}, 49 - 1 \times 10^{-9}]$		4.196611619528941E+01	44	1044*	22	29	14	20	105	23	12	16	12	105	13	12	12	58	12	11	11	13
8			$[49 + 1 \times 10^{-9}, 64 - 1 \times 10^{-9}]$		5.50533958014309E+01	44	1044*	22	29	22	16	105	26	13	16	105	13	12	12	60	12	49	13	13	
9			$[64 + 1 \times 10^{-9}, 81 - 1 \times 10^{-9}]$		7.19856558658780E+01	44	1044*	84	29	18	18	106	80	21	28	204*	106	13	12	12	62	12	12	11	13
10			$[81 + 1 \times 10^{-9}, 100 - 1 \times 10^{-9}]$		9.00088653916667E+01	43	1043*	84	64	16	24	106	80	16	16	104*	106	13	12	11	62	12	12	12	13
11			$[100 + 1 \times 10^{-9}, 121 - 1 \times 10^{-9}]$		1.100265327483302E+02	43	1043*	20	34	17	14	106	23	13	14	12	106	12	11	11	61	11	11	11	13
12		40	$[-9, 31]$		0.00000000000000E+00	58	1058*	30	56	18	4	50	38	18	20	20	176	22	16	16	132	18	14	14	14
12	$-nxe^{-x}$	100	$[-9, 31]$		0.00000000000000E+00	58	1058*	30	56	20	4	50	38	18	20	20	176	22	16	16	132	18	14	14	14
12		200	$[-9, 31]$		0.00000000000000E+00	58	1058*	30	56	20	4	50	38	18	20	20	176	22	16	16	132	18	14	14	14
13	$x^3 - 2x - x + 3$		$[-3, 2]$		-2.1083040275537E+00	47	46	18	40	18	17	15	35	11	18	36	13	12	12	12	13	12	62	12	12
14	$e^{-x} - x - \sin(x)$		$[0, 2]$		3.54631043750253E-01	48	16	18	28	14	14	10	20	12	8	8	8	8	8	8	8	8	7	8	8
15		4	$[0, 5]$		6.687403049764220E-01	49	1049*	20	25	16	18	19	29	49	16	18	17	15	14	14	106	14	14	13	12
15		6	$[0, 5]$		7.64724913317300E-01	48	1018*	28	34	21	14	26	32	49	18	20	25	18	16	16	1048*	16	18	13	13
15	$x^n - 0.2$	8	$[0, 5]$		8.177654339579425E-01	48	1018*	26	27	19	14	32	35	49	20	24	31	18	17	17	1048*	17	21	14	13
15		10	$[0, 5]$		8.513399225078746E-01	48	1018*	28	33	19	16	37	35	49	20	22	36	17	16	16	1048*	17	24	14	13
15		12	$[0, 5]$		8.7485272211678E-01	48	1018*	28	33	21	16	42	35	49	20	18	40	17	17	17	1048*	17	26	15	14
16		4	$[0, 5]$		1.00000000000000E+00	48	1018*	24	34	18	16	19	32	13	14	16	17	13	15	15	53	14	13	13	12
16		6	$[0, 5]$		1.00000000000000E+00	48	1018*	24	33	17	12	24	32	14	16	16	22	13	15	15	906	15	16	13	12
16	$x^n - 1$	8	$[0, 5]$		1.00000000000000E+00	48	1018*	22	48	17	16	29	32	20	18	18	29	15	15	15	1048*	14	19	13	13
16		10	$[0, 5]$		1.00000000000000E+00	48	1018*	18	44	19	16	36	32	25	18	18	33	18	16	16	1048*	15	22	14	13
16		12	$[0, 5]$		1.00000000000000E+00	48	1018*	28	23	16	40	23	16	40	23	20	20	18	17	17	1048*	17	24	14	13
17		8	$[-0.95, 4.05]$		1.00000000000000E+00	48	1018*	28	59	23	12	33	32	38	18	16	30	22	15	15	37	14	15	14	12
17	$x^{n^2} - 1$	10	$[-0.95, 4.05]$		1.00000000000000E+00	48	1018*	28	44	22	16	39	35	22	18	34	22	16	16	16	49	15	16	14	12
17		12	$[-0.95, 4.05]$		1.00000000000000E+00	48	1018*	28	43	22	16	44	35	38	20	20	37	22	16	16	72	15	17	14	13
17		14	$[-0.95, 4.05]$		1.00000000000000E+00	48	1018*	28	45	24	16	49	35	38	20	20	40	24	17	17	129	16	19	14	14
18	$\sin x - 0.5$		$[0, 1.5]$		5.2358757556288E-01	47	17	18	25	14	14	11	23	10	12	8	8	9	9	9	9	8	9	8	
19		1	$[0, 1]$		4.224777096112367E-01	20	18	17	12	10	17	9	12	12	9	10	8	9	7	7	7	6	7	7	12
19		2	$[0, 1]$		3.06699410483207E-01	47	28	20	29	14	10	11	26	11	12	10	9	8	9	8	8	8	8	9	8
19		3	$[0, 1]$		2.237054576546630E-01	48	32	20	21	14	10	23	11	12	10	10	9	9	9	9	9	9	9	10	9
19		4	$[0, 1]$		1.717191475195084E-01	48	33	18	24	15	12	12	29	13	12	11	10	10	10	10	10	10	10	10	10
19		5	$[0, 1]$		1.382571550568241E-01	49	33	16	20	14	12	12	29	13	12	12	10	10	10	10	10	10	10	10	10
19	$2xe^{-n} - 2e^{-nx} + 1$	10	$[0, 1]$		6.931408867972348E-02	50	35	25	18	14	12	13	32	14	14	12	11	11	11	11	12	11	11	11	9
19		20	$[0, 1]$		3.46575923408535E-02	41	34	24	24	17	12	13	32	15	14	14	11	12	12	12	13	12	12	12	12
19		40	$[0, 1]$		1.73286795139863E-02	52	36	18	31	18	14	14	35	16	14	14	13	13	13	13	15	13	13	13	13
19		60	$[0, 1]$		1.15524530093324E-02	52	37	24	28	19	14	14	35	23	16	16	13	13	14	14	16	13	13	15	14
19		80	$[0, 1]$		8.66433975699316E-03	52	37	26	29	19	14	15	35	17	16	16	13	13	14	14	15	14	14	14	14
19		100	$[0, 1]$		6.93147180559641E-03	52	38	28	29	18	14	15	35	16	16	16	13	13	14	14	16	14	14	15	14
20		1	$[0, 1]$		3.84025154082190E-02	45	17	15	14	17	12	13	32	10	4	9	8	8	8	8	9	5	9	10	9
20	$(1 + (1 - n)^2)x - (1 - nx)^2$	10	$[0, 1]$		9.90009998040499E-03	52	12	12	21	17	16	9	20	10	8	6	9	7	8	8	8	8	6	9	8
20		20	$[0, 1]$		2.49375003960212E-03	52	10	10	27	17	18	8	20	9	6	6	9	7	8	8	7	8	9	9	9
21		2	$[0, 1.01]$		5.00000000000000E-01	47	3	4	5	4	3	5	8	4	4	3	3	3	3	3	3	3	4	4	4
21	$x^2 - (1 - x)^n$	20	$[0, 1.01]$		3.459548158482420E-01	47	44	20	26	14	12	9	23	13	12	12	9	10	10	8	9	9	10	9	8
21		10	$[0, 1.01]$		2.45122337533072E-01	48	91	20	23	16	14	11	29	16	14	14	11	10	10	10	10	10	12	10	10
21		20	$[0, 1.01]$		1.95347625365666E-01	48	101	24	27	16	14	11	26	14	18	12	12	12	12	11	11	14	11	11	13
21		20	$[0, 1.01]$		1.649209572764410E-01	48	191	24	31	14	14	12	32	47	16	20	12	11	12	12	12	12	13	12	12
22		1	$[0, 1]$		2.755080409994844E-01	48	32	20	25	16	10	11	26	11	12	10	9	8	9	9	8	8	9	9	9
22		2	$[0, 1]$		1.377540204997422E-01	49	33	18	20	14	10	12	29	13	12	12	10	10	10	10	10	9	9	10	10
22		4	$[0, 1]$		1.030528377815644E-02	52	12	14	22	12	12	10	23	10	12	12	9	10	8	8	9	8	8	9	8
22	$(1 + (1 - n)^4)x - (1 - nx)^4$	5	$[0, 1]$		3.6171089704063E-03	52	10	12	33	10	10	9	20	9	10	10	8	10	7	7	8	7	8	8	9
22		8	$[0, 1]$		4.10872914963954E-04	8	10	36	12	12	12	8	17	16	12	12	8	7	8	7	8	7	10	8	8
22		15	$[0, 1]$		2.508957589207063E-05	52	7	10	30	12	12	8	17	8	12	12	7	8	6	6	7	6	9	7	7
22		20	$[0, 1]$		7.668595122185337E-06	52	7	8	27	12	12	8	14	8	12	12	7	8	6	6	7	6	8	7	7
23		1	$[0, 1]$		4.010581375415471E-01	47	22	18	17	12	10	10	20	8	10	9	8	7</							